

Machine learning for sample-based quantum diagonalization: generative configuration recovery and the classical-simulability frontier

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Abstract

Sample-based quantum diagonalization (SQD) — and the equivalent quantum-selected configuration interaction (QSCI) — have, in the span of two years, become the pragmatic centre of gravity of quantum-computational chemistry in the pre-fault-tolerant era: a quantum processor samples electronic configurations from an approximate ground state, and the many-electron Hamiltonian is diagonalized classically in the resulting determinant subspace. The efficiency of this hybrid loop is governed entirely by *which* configurations enter the subspace — a selection problem that is the natural entry point for machine learning, and one that a coupon-collector bottleneck makes acute. A diverse ecosystem of generative and learned methods has grown around this step: restricted Boltzmann machines for configuration recovery, autoregressive and transformer neural quantum states embedded in the diagonalization loop, neural-network classifiers and learning-to-rank selectors, and generative circuit design. Here we critically review this ecosystem. We organize the methods by the object they generate and the importance signal they exploit — a taxonomy that also exposes a conspicuous methodological gap: the absence of any reward-proportional generative-flow-network proposer built for exactly the tail-discovery problem the coupon-collector law poses — and we situate the whole against the classical selected-configuration-interaction methods that are its true benchmark. We then confront the central, contested question the enterprise turns on: whether the quantum sampler, with or without machine-learning augmentation, actually improves on classical selected configuration interaction. The honest current answer is a clean negative, and it is worth stating without flinching: in every published same-active-space comparison we are aware of — a small and largely critic-assembled set in which every comparison points the same direction — classical heat-bath and semistochastic selected-CI match or beat the quantum-sampled subspace; random determinants passed through the classical recovery loop reproduce the accuracy; and in 2026 the single-layer local unitary cluster Jastrow circuit used in the flagship experiments was shown to admit polynomial-time classical energy estimation — weak simulation, not strong bitstring simulation — reproducing the 77-qubit iron-sulfur result at lower energy than the noisy device, though whether deep multi-layer circuits evade simulation remains open. From this we distil a benchmarking standard — exact-ground-truth controls, strong classical *and* dequantization baselines, full cost accounting, and multi-seed error bars — that we argue any future claim in the field must meet, and against which we hold even our own illustrative computations. We then read the negative result not as a dead end but as a compass — a conditional map, not a universal verdict. Its needle is set by a single structural observation: on molecular systems the projected-Hamiltonian structure that makes an SQD subspace classically *verifiable* is empirically the same structure that furnishes a cheap classical importance heuristic with which to *construct* it — so a quantum or generative edge can survive only where that coincidence breaks, in the noise-limited configuration-recovery regime and in genuinely multireference chemistry, where the cheap classical importance signal degrades. (Verifiability does not in general imply cheap constructibility — that is the P-versus-NP gap — which is why the compass is a conditional map and not a theorem.) Beyond the SQD loop we flag one further horizon, the quantum-data-native task of learning from experiments, whose classical sample-complexity lower bound is an unconditional theorem rather than a dequantizable conjecture; its bridge to useful chemistry we frame as an open problem, not a present advantage of the methods reviewed here.

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1. Introduction

The determination of the ground-state energy of a many-electron Hamiltonian is among the oldest and most consequential problems in the physical sciences, underlying our predictive understanding of chemical reactivity, catalysis, and correlated materials. Its difficulty is structural: the dimension of the full configuration-interaction (FCI) space grows combinatorially with the number of electrons and orbitals, and the exact problem is, in the worst case, QMA-complete [Kempe–Kitaev–Regev 2006]. The last half-century of quantum chemistry can be read as a sequence of increasingly sophisticated strategies for evading this exponential wall without paying its full cost — coupled-cluster expansions, the density-matrix renormalization group (DMRG), auxiliary-field quantum Monte Carlo (AFQMC), and, most directly relevant here, *selected* configuration interaction (sCI), which builds a compact variational subspace by retaining only the determinants that matter and correcting the remainder perturbatively [Huron et al. 1973; Holmes et al. 2016; Sharma et al. 2017; Tubman et al. 2016].

The advent of programmable quantum processors reframed the same problem. If the exponential difficulty of electronic structure is a quantum difficulty, perhaps a quantum device can help. The first decade of this hope was dominated by the variational quantum eigensolver (VQE) [Peruzzo et al. 2014], in which a parameterized circuit is optimized to minimize an energy expectation value. That programme has, for the purpose of demonstrating a scaling advantage, largely run its course: barren plateaus render deep expressive circuits untrainable [McClean et al. 2018; Larocca et al. 2025], the measurement cost of estimating chemical-accuracy energies is prohibitive, and — most damningly — the structural conditions that let a variational circuit *avoid* barren plateaus tend also to render it classically simulable [Cerezo et al. 2025]. Into the vacuum left by VQE stepped a conceptually different, and strikingly effective, idea.

Sample-based quantum diagonalization. Rather than optimize a circuit to *be* the ground state, one uses a shallow, noise-tolerant circuit only to *sample* the electronic configurations — computational-basis bitstrings, i.e. Slater determinants — that carry appreciable ground-state weight, and then diagonalizes the Hamiltonian classically in the sampled subspace on high-performance hardware. Introduced as quantum-selected configuration interaction (QSCI) by Kanno and co-workers [Kanno et al. 2023] and scaled to quantum-centric supercomputing as SQD by an IBM–RIKEN collaboration [Robledo-Moreno et al. 2025], the method inherits the variational guarantee of sCI, sidesteps the optimization and measurement pathologies of VQE, and is remarkably robust to hardware noise through a self-consistent configuration-recovery step that repairs symmetry-broken samples. Within two years it has been demonstrated on iron–sulfur clusters at 77 qubits with the entire Fugaku supercomputer in the loop [Robledo-Moreno et al. 2025; Shirakawa et al. 2025], extended to excited states, embeddings, and periodic systems, and pushed — through fragment-based workflows — to a 12,635-atom protein–ligand complex [Wang et al. 2026]. SQD is, today, the pragmatic centre of gravity of pre-fault-tolerant quantum chemistry.

The machine-learning entry point. The accuracy of the SQD loop is determined entirely by which configurations populate the subspace, and this is where the method is most vulnerable and most interesting. The relevant determinants are heavy-tailed in amplitude; discovering the rare, chemically decisive ones by repeated sampling is a coupon-collector problem, in which the marginal cost of each newly discovered configuration grows sharply [Reinholdt et al. 2025]. The configuration-recovery step — a probabilistic repair of noise-corrupted samples toward an estimated occupation profile — is a statistical denoising problem. Both are natural targets for machine learning, and the field has responded with striking speed. In the two years since QSCI, generative and learned methods have been proposed for every stage of the loop: restricted Boltzmann machines that learn and resample the distribution of dominant determinants [Patra et al. 2025; 2026]; autoregressive and transformer neural quantum states that propose determinants inside the sample–diagonalize–update loop [Chang et al. 2026; Solanki et al. 2026; Shang et al. 2025]; neural-network classifiers and learning-to-rank models that predict determinant importance [Zeni et al. 2026; Zhang et al. 2025; Nie et al. 2026]; and generative circuit design that learns the sampling ansatz itself [Kemmu et al.

al. 2026]. This ecosystem has, to our knowledge, never been reviewed, and it is growing faster than any single group can track.

The question the field turns on. Beneath the methodological activity lies a single, contested, and increasingly uncomfortable question: does any of this actually beat classical selected configuration interaction? SQD *is*, after all, a quantum-sampled sCI, and its classical competitors — semistochastic heat-bath CI (SHCI), the perturbatively-selected CIPSI, adaptive sampling CI (ASCI), DMRG, and AFQMC — are mature, fast, and run on a laptop. The critical literature has grown in tandem with the method. Reinholdt and co-workers showed that, even granting SQD a perfect noiseless sampler, its expansions are an order of magnitude less compact than heat-bath CI on the very systems used to showcase it [Reinholdt et al. 2025]; Gaberle and Jattana found that for strongly correlated lattice models the required configuration count grows exponentially *even with an optimal sampler* [Gaberle–Jattana 2026]; Vaquero-Sabater and co-workers demonstrated that random determinants passed through the classical recovery loop reproduce the accuracy, implying that the classical recovery, not the quantum sampler, is doing the work [Vaquero-Sabater et al. 2026]; and, most strikingly, Belagali and co-workers showed in 2026 that the single-layer local unitary cluster Jastrow circuit SQD samples is classically simulable in polynomial time, reproducing the flagship 77-qubit experiment on a laptop at *lower* energy [Belagali et al. 2026]. The pro-quantum case — worst-case instantaneous-quantum-polynomial (IQP) hardness of the sampling circuits [Hafid et al. 2025], a contested 49-qubit advantage claim on sparse-ground-state Hamiltonians [Kirby et al. 2026] — remains alive but narrow.

This review. We take the position that a field cannot mature while its central claim is unsettled, and that the honest place to stand is neither uncritical enthusiasm nor dismissal, but a rigorous map of what is known. We make four contributions. First (§2–§3), we give a structured account of the SQD/QSCI method family and the configuration-recovery bottleneck that makes it a machine-learning problem. Second (§4), we provide the first taxonomy of the ML and generative methods proposed for determinant and subspace selection, organized by the object they generate and the importance signal they exploit, and we identify the conspicuous methodological gap — that generative flow networks, purpose-built for diverse reward-proportional sampling over exponential discrete spaces, have not yet been applied to this problem. Third (§5), we confront the quantum-advantage question directly, assembling the classical baselines, the benzene-benchmark accuracy standard, the critique literature, and the 2026 dequantization results into an honest verdict. Fourth (§6), we propose a set of benchmarking standards that we contend any future claim in this field must meet; and (§7) we argue that the negative result is a compass, identifying the regimes — noise-limited recovery, genuine multireference character, and the distinct quantum-data-native task of learning from experiments — where generative and quantum methods retain a defensible or, in one case, provable advantage. Throughout, we distinguish sharply between what is proven, what is demonstrated, and what is hoped.

2. Sample-based quantum diagonalization: the method and its family

Scope and method. This review surveys the sample-based quantum diagonalization literature indexed on arXiv (quant-ph, physics.chem-ph) and in the major chemistry and quantum-information journals, together with the community bibliography *Awesome-QSCI*, from the introduction of QSCI in early 2023 to mid-2026. We include the SQD/QSCI method family, the machine-learning and generative methods proposed for determinant or subspace selection, the classical selected-configuration-interaction competitors that constitute the benchmark, and the critique and dequantization literature; we exclude general variational-quantum-eigensolver work except where it bears directly on the sampling step. Because the field is moving on a scale of weeks and much of the primary literature is preprint, we date every claim explicitly and flag its standing — established, demonstrated, or merely hoped.

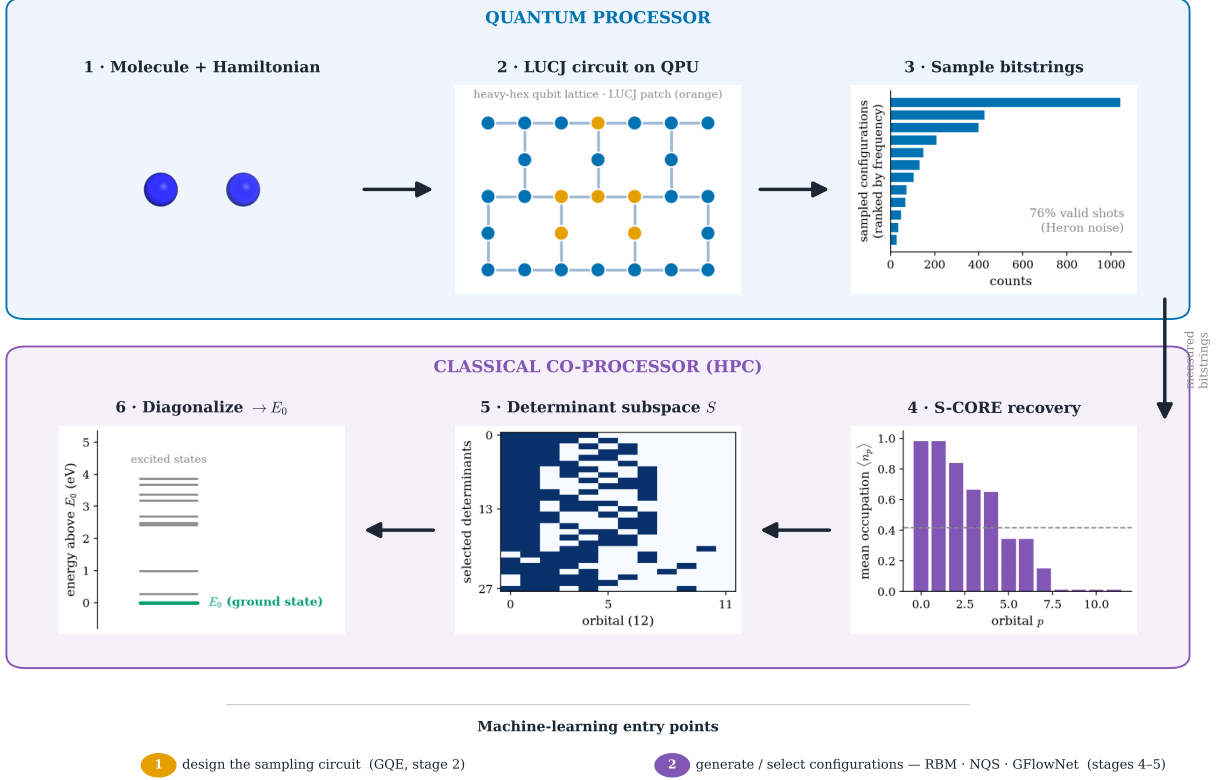


Figure 1. The sample-based quantum diagonalization workflow, with real data at every stage (N_2 , CAS(10e,12o), exact-FCI-verifiable). A shallow LUCJ circuit on the quantum processor (orbital rotations $U(\theta)$ interleaved with diagonal-Coulomb Jastrow gates) is sampled in the computational basis; here 76% of shots survive particle-number post-selection at Heron-calibrated noise. The classical co-processor repairs the broken samples by occupation-weighted S-CORE, assembles the recovered determinants into a subspace $\mathcal{S}$ (each row of panel 5 is one selected configuration’s orbital occupation), and diagonalizes the projected Hamiltonian, whose energy is a variational upper bound converging to the exact FCI value (panel 6). Machine learning augments either the sampler (①, generative circuit design) or the generation/selection of configurations (②). Every panel is real output of the accompanying calculations.

2.1 The core loop

The object shared by all methods in this review is a hybrid quantum–classical loop with four stages. (i) A parameterized quantum circuit prepares an approximate ground state. In the flagship demonstrations this is the *local unitary cluster Jastrow* (LUCJ) ansatz [Motta et al. 2023], a hardware-efficient truncation of unitary coupled cluster that alternates orbital rotations with diagonal Coulomb (Jastrow) factors and restricts entangling operations to a physically motivated local connectivity, making it native to heavy-hex superconducting lattices and initializable from classical CCSD amplitudes. (ii) The circuit is measured in the computational basis; under the Jordan–Wigner mapping each length- $2M$ bitstring (for M spatial orbitals) is an electronic configuration, split into an α and a β half. Each measured bitstring is thus a *proposed* Slater determinant, and the collection of distinct bitstrings, with their frequencies, is the raw material the rest of the pipeline processes. (iii) In SQD, noise corrupts a large fraction of these bitstrings to the wrong particle number or spin; a *self-consistent configuration recovery* (S-CORE) step repairs them, flipping spin-orbitals toward the mean occupation of the current best eigenvector and iterating to self-consistency [Robledo-Moreno et al. 2025]; we make the mechanism precise below. (iv) The Hamiltonian is projected onto the recovered determinant subspace and diagonalized by a sparse (Davidson) eigensolver, distributed across high-performance classical hardware; the result is a variational upper bound and a sparse ground-state wavefunction.

Formally, the electronic Hamiltonian in second quantization is

$$\hat{H} = \sum_{pq} h_{pq} \hat{a}_p^\dagger \hat{a}_q + \frac{1}{2} \sum_{pqrs} (pq|rs) \hat{a}_p^\dagger \hat{a}_r^\dagger \hat{a}_s \hat{a}_q + E_{\text{nuc}}, \quad (1)$$

whose exact diagonalization in the full configuration-interaction (FCI) basis of M active spatial orbitals with (n_α, n_β) electrons has dimension $\dim \mathcal{H}_{\text{FCI}} = \binom{M}{n_\alpha} \binom{M}{n_\beta}$, growing combinatorially. SQD replaces $\mathcal{H}_{\text{FCI}}$ by a sampled subspace $\mathcal{S} = \text{span}\{|D_x\rangle\}_{x \in \mathcal{S}}$ of Slater determinants and solves the projected eigenproblem for the projector $\hat{P}_{\mathcal{S}} = \sum_{x \in \mathcal{S}} |D_x\rangle \langle D_x|$,

$$(\hat{P}_{\mathcal{S}} \hat{H} \hat{P}_{\mathcal{S}}) \mathbf{c} = E_{\mathcal{S}} \mathbf{c}, \quad E_{\mathcal{S}} \geq E_{\text{FCI}}, \quad (2)$$

the inequality (the Hylleraas–Undheim–MacDonald theorem) being the variational guarantee that makes every SQD energy a rigorous upper bound that improves monotonically as good determinants are added. This guarantee is a double-edged asset, and the edge it cuts against is the theme of §5–§7: the same projected-Hamiltonian structure that lets one *certify* an SQD energy from the classical side is what lets one *construct*, screen, and — for the shallow circuits actually run — *simulate* the subspace classically. The variational principle that grounds the method’s appeal is thus also the doorway through which its classical competitors enter. The self-consistent recovery (S-CORE) closes the loop: from the current subspace eigenvector it forms the mean spin-orbital occupations $\langle n_p \rangle = \sum_{x \in \mathcal{S}} |c_x|^2 [x]_p$ and repairs each symmetry-broken bitstring by flipping spin-orbitals with probability increasing in the deviation $|[x]_p - \langle n_p \rangle|$, iterating recovery $\rightarrow$ diagonalization $\rightarrow$ occupation-update to self-consistency. Because the whole quantitative analysis of this review lives in an active space small enough for exact FCI — N_2 and H_2O in CAS(10e,12o)/(8e,12o), whose active-space orbitals are shown in Fig. 2 — every energy we quote below is an *error against exact truth*, not against another approximation.

Two lineages coined the idea nearly simultaneously and use different names for it. *Quantum-selected configuration interaction* (QSCI) was introduced by Kanno and co-workers [Kanno et al. 2023], now published in *Physical Review Research*, and framed the method as “the quantum computer selects the subspace, the classical computer diagonalizes.” *Sample-based quantum diagonalization* (SQD) is the IBM–RIKEN realization at quantum-centric-supercomputing scale [Robledo-Moreno et al. 2025], adding the self-consistent recovery loop and the tight coupling to an HPC back end. We use SQD and QSCI interchangeably, noting the recovery step as the principal methodological difference and the natural home of the machine-learning contributions reviewed in §4. Crucially, because the final step is an exact diagonalization in a fixed subspace, the energy is variational: it can only be an upper bound to the true ground state, and it improves monotonically as good determinants are added — a property that both grounds the method’s appeal and, as §5 will show, sets the exact bar its classical competitors must be measured against.

2.2 A taxonomy of variants

In two years the SQD/QSCI family has branched into some fifty distinct variants; we organize them by the stage of the loop they modify. *Input-state variants* replace the LUCJ sampler: ADAPT-QSCI grows the state one Pauli generator at a time without a variational optimization [Nakagawa et al. 2024]; time-evolved QSCI (TE-QSCI) and Hamiltonian-simulation QSCI (HSB-QSCI) generate multi-order excitations from real-time evolution rather than circuit optimization [Mikkelsen–Nakagawa 2025; Sugisaki et al. 2025]; and generative circuit design learns the sampler itself (§4.5). *Krylov variants* — sample-based Krylov quantum diagonalization (SKQD) and its randomized, provable-convergence descendant SqDRIFT — build a Krylov subspace from sampled time-evolved states, importing phase-estimation-like convergence guarantees at low depth [Yu et al. 2025; Piccinelli et al. 2025]. *Recovery and selection variants* attack the coupon-collector bottleneck directly: amplitude amplification of unseen bitstrings [Stockinger et al. 2026], active-sampling by an

The molecular systems, their active-space orbital-energy structure, and the frontier orbitals

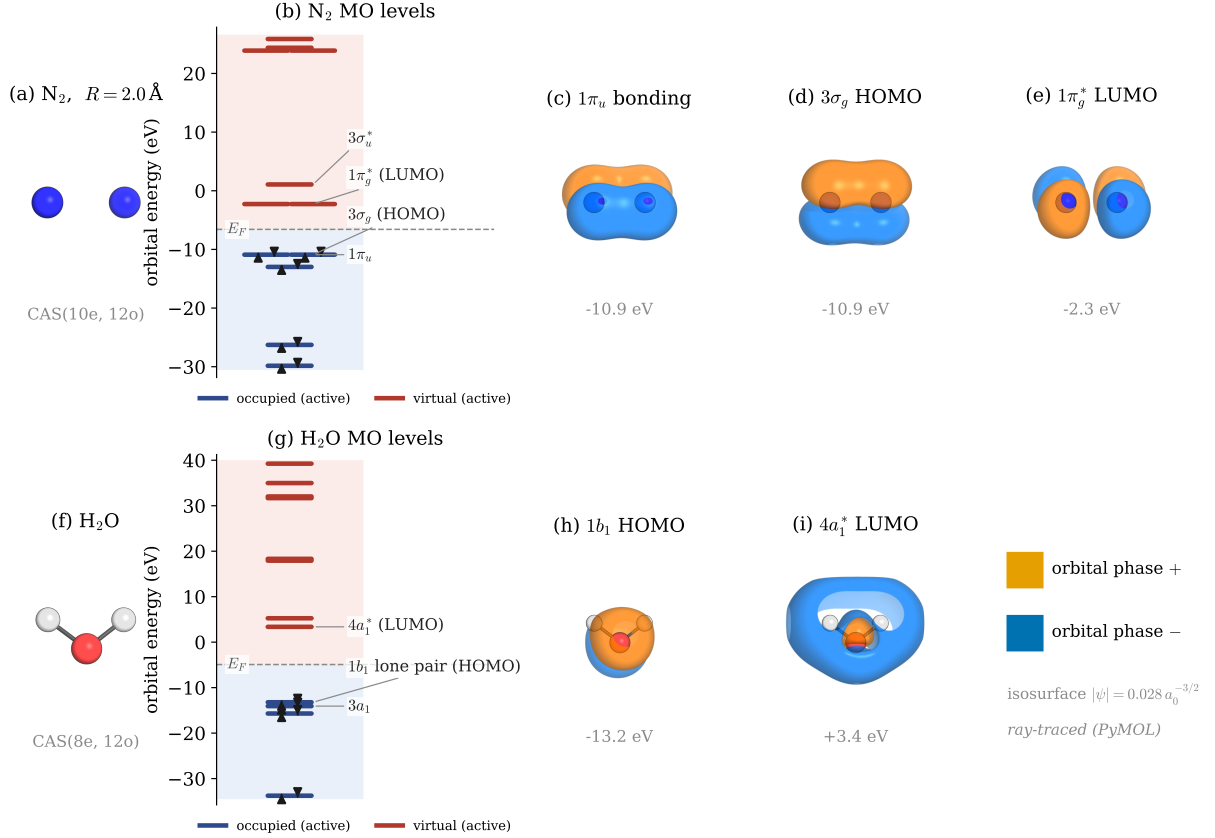


Figure 2. The molecular systems, their active-space orbital-energy structure, and the frontier orbitals. (a,f) Ray-traced ball-and-stick geometries of stretched N_2 ($R = 2.0 \text{ \AA}$, CAS(10e,12o)) and H_2O (CAS(8e,12o)). (b,g) Molecular-orbital energy-level diagrams (restricted Hartree-Fock, eV): occupied active levels (blue) carry their two electrons as up/down arrows, virtual active levels are red, the shaded band is the complete active space, and E_F marks the HOMO-LUMO midpoint; the deep $1s$ core is frozen and off-scale. (c-e) Ray-traced (PyMOL) isosurfaces of the N_2 $1\pi_u$ bonding, $3\sigma_g$ HOMO, and $1\pi_g^*$ LUMO orbitals, with their energies; the near-degenerate π/π^* manifold is what makes stretched N_2 strongly multireference. (h,i) The H_2O $1b_1$ lone-pair HOMO and $4a_1^*$ LUMO. Isosurfaces at $|\psi| = 0.028 a_0^{-3/2}$; orbital phases orange/blue.

Epstein-Nesbet acquisition function [Miura et al. 2026], and — the focus of §4 — machine-learned recovery. *Qubit-reduced variants* halve the register through half-qubit encodings, seniority-zero sampling, entanglement forging, or neural fermionic compression [McFarthing et al. 2026; Yoshida et al. 2026; Smith et al. 2025; Chen et al. 2025]. *Embedding variants* reach extended systems by using SQD as the impurity or fragment solver inside density-matrix embedding, localized-active-space SCF, or DFT embedding [Shajan et al. 2025; Wang et al. 2026]. *Hybrid post-processing variants* recover dynamical correlation missed by the finite subspace, using the SQD wavefunction as an AFQMC trial, a tailored-CC reference, or a multireference-PT2 anchor [Danilov et al. 2025; Erhart et al. 2025]. A full annotated tabulation, with citations, systems, and largest scale, is given in the accompanying dataset.

2.3 Hardware demonstrations and honest scale

The method’s hardware trajectory is genuinely impressive and worth stating precisely, because it is often conflated with an advantage claim it does not make. The founding demonstration reached N_2 at 58 qubits and the iron-sulfur clusters $[2\text{Fe-2S}]$ and $[4\text{Fe-4S}]$ at 45 and 77 qubits — a (54e,36o) active space, the largest quantum-computed at the time — with classical diagonalization distributed across

Table 1. The SQD/QSCI method family (curated; the full annotated list is maintained in the state-of-the-art dataset). “Largest scale” states hardware qubit counts or active spaces as reported.

Category	Method	Key idea	Ref.	Largest scale
Core	QSCI	Quantum samples the subspace; classical diagonalization	[Kanno et al. 2023]	8 qubits
	SQD + S-CORE	Self-consistent configuration recovery; quantum-centric supercomputing	[Robledo-Moreno et al. 2025]	77 q / (54e,36o)
Sampler/input	ADAPT-QSCI	Iterative input-state growth, no VQE	[Nakagawa et al. 2024]	small molecules
	TE-/HSB-QSCI	Time-evolved sampling state	[Mikkelsen–Nakagawa 2025]	36 q (carbyne)
	Generative-QSCI	Transformer (GQE) designs the sampling circuit	[Kemmoku et al. 2026]	32 q (N ₂)
Krylov	SKQD / SqDRIFT	Sampled Krylov subspace; provable convergence	[Yu et al. 2025; Piccinelli et al. 2025]	49 q
Recovery/select	SQD-AA	Amplitude-amplify unseen bitstrings (quadratic gain)	—	model systems
	ML recovery	Learned generation/classification (see Table 2)	—	see Table 2
Excited/orbital	ext-/oo-SQD	Extra projection for excited states; orbital optimization	[Barison et al. 2025]	CH ₂ 52 q
Symmetry/compact	Compact-QSCI	Space-group symmetry; time-evolved compact subspaces	[Weaving et al. 2025; Nogaki et al. 2026]	SiH ₄ 42 q
Qubit-reduced	HCI-HSQD	Half-qubit; seniority-zero; neural compression	[McFarthing et al. 2026]	Fe-S (54e,36o)
Embedding	DMET-/LAS-SQD	SQD as fragment / impurity solver	[Wang et al. 2026]	protein 12,635 atoms
Hybrid post-proc.	AFQMC-SQD; PT2	Recover dynamical correlation on the SQD reference	—	N ₂ ; [2Fe-2S]

up to 6,400 nodes of Fugaku [Robledo-Moreno et al. 2025]. A full-scale closed-loop follow-up used the entire 152,064-node Fugaku machine with an on-premises Heron processor and improved the [4Fe-4S] energy by 278 mHa over the 2024 result [Shirakawa et al. 2025]. Fragment-based embeddings then pushed the workflow, in 2025–2026, from a $\sim$ 300-atom miniprotein to a 12,635-atom protein–ligand complex [Wang et al. 2026]. These are landmark demonstrations of *scale beyond exact diagonalization*. They are not, and are not claimed by their authors to be, demonstrations of an advantage over the best classical methods — a distinction §5 makes central.

2.4 Software and ecosystem

The method is unusually well tooled for its age, which partly explains its rapid uptake. IBM maintains `qiskit-addon-sqd` (with an HPC C++ variant) and `ffsim` for fermionic-circuit simulation and LUCJ construction; QunaSys maintains `quri-parts-qsci` within the QURI SDK and, notably, a continuously updated community bibliography, *Awesome-QSCI*, that functions as a living review of a field moving too fast for the conventional publication cycle. NVIDIA’s CUDA-Q documents SKQD as a reference application and provides GPU primitives for the classical diagonalization back end. This tooling maturity is a double-edged asset: it lowers the barrier to method development — including the machine-learning methods of §4 — but it also lowers the barrier to producing demonstrations that are not benchmarked against the strong classical baselines

of §5.

3. The configuration-recovery bottleneck: the machine-learning entry point

3.1 A coupon-collector law

The efficiency of the entire loop reduces to a single question: how many distinct, *important* determinants can be placed in the subspace, and at what cost? The answer is governed by the amplitude distribution of the exact ground state, which is heavy-tailed. In stretched N_2 in a (10e,12o) active space, for instance, 90% of the exact FCI weight is carried by roughly 1.5% of the single-spin strings (12 of the 792), with per-configuration weights spanning many orders of magnitude. Sampling from such a distribution to *discover* the low-amplitude but chemically decisive determinants is a coupon-collector problem with non-uniform probabilities: as sampling proceeds, an overwhelming fraction of draws re-select already-seen configurations, and the number of newly discovered determinants per shot collapses. Reinholdt and co-workers made this quantitative and damaging: for N_2 the discovery rate falls from ~ 0.1 determinants per shot at 10^3 shots to $\sim 5 \times 10^{-4}$ at 10^9 , and reaching micro-hartree precision extrapolates to on the order of 10^{14} samples [Reinholdt et al. 2025]. This is the bottleneck that machine learning is invoked to break: if a learned model can *generate* the important tail rather than wait to sample it, the coupon-collector wall might be circumvented. Whether it can — and whether doing so beats simply running a classical selected-CI heuristic — is the question of §4 and §5.

The effect is exact and computable. Writing the marginal weight of a single-spin string i as $w_\alpha(i) = \sum_\beta |c_{i\beta}|^2$ and its sampling probability as p_i , the expected number of shots required to first observe every string in a target set $\mathcal{T}$ is the coupon-collector sum

$$\mathbb{E}[\text{shots to collect } \mathcal{T}] \simeq \sum_{i \in \mathcal{T}} \frac{1}{p_i} \xrightarrow{p_i \rightarrow 0} \infty, \quad (3)$$

which upper-bounds the exact expectation (the leading term of an inclusion–exclusion series) and diverges as the heavy tail is approached. Fig. 3 computes this directly from the exact FCI wavefunction of N_2 CAS(10e,12o): the same handful of dominant strings carry almost all the weight, but the correlation-energy tail that decides chemical accuracy is spread over hundreds of rare strings — precisely the configurations whose p_i make the sum above large, and precisely where a learned generator is supposed to earn its keep.

3.2 Recovery as denoising, and the peculiar role of noise

On real hardware the bottleneck is compounded by noise, and here the picture is subtle in a way that is central to the field’s honest self-assessment. Under readout and gate noise, the majority of measured bitstrings violate particle-number conservation and would be discarded by naive post-selection — for N_2 in a large active space on hardware, only about 0.17% of raw samples survive [Vaquero-Sabater et al. 2026]. The self-consistent configuration recovery step is precisely a statistical imputation model that rescues these corrupted samples, and it is the true workhorse of SQD’s noise tolerance. Strikingly, Vaquero-Sabater and co-workers showed that noise can *help*: by scattering samples beyond the biased support of the ideal ansatz, moderate noise broadens Hilbert-space exploration, and recovery starting from *uniformly random* bitstrings reaches accurate energies within a couple of iterations. The implication — that the classical recovery loop, not the quantum sampler, is responsible for the accuracy — is one of the sharpest results in the literature and is examined in §5. For the present section the point is narrower: the recovery step is a learned/statistical model, and it is the natural site for the generative methods of §4, whose promise is to replace an occupancy-weighted heuristic repair with a model that has learned the structure of the important subspace. Fig. 4 makes the mechanism concrete on backend-calibrated noise: as the device-noise scale grows

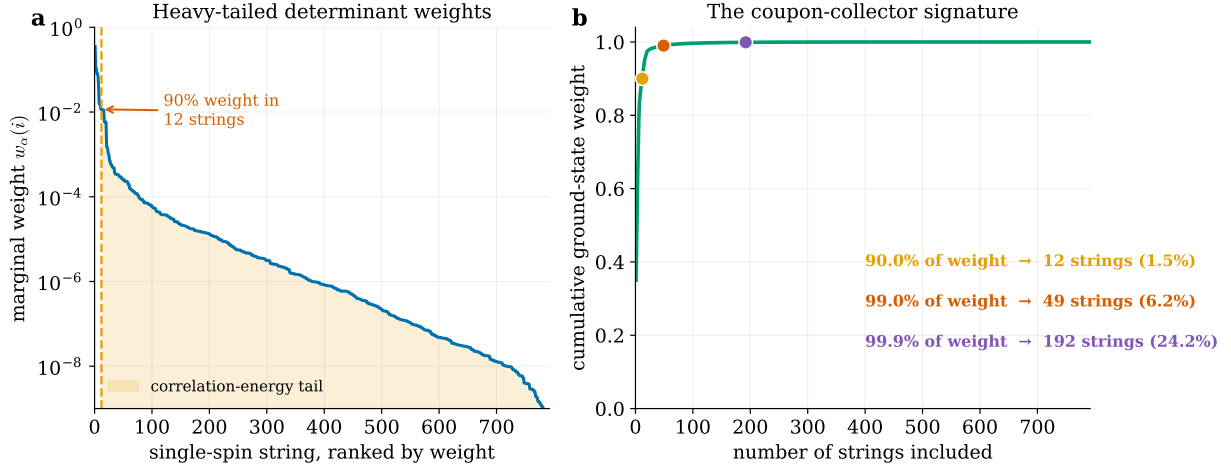


Figure 3. The coupon-collector bottleneck, computed exactly. N_2 CAS(10e,12o), from the exact FCI wavefunction. *Left:* the marginal single-spin-string weight $w_\alpha(i) = \sum_\beta |c_{i\beta}|^2$ falls over three orders of magnitude. *Right:* 90% of the ground-state weight is carried by 12 of 792 strings (1.5%), but the correlation-energy tail is spread over hundreds of rare strings — the configurations a sampler must pay to discover.

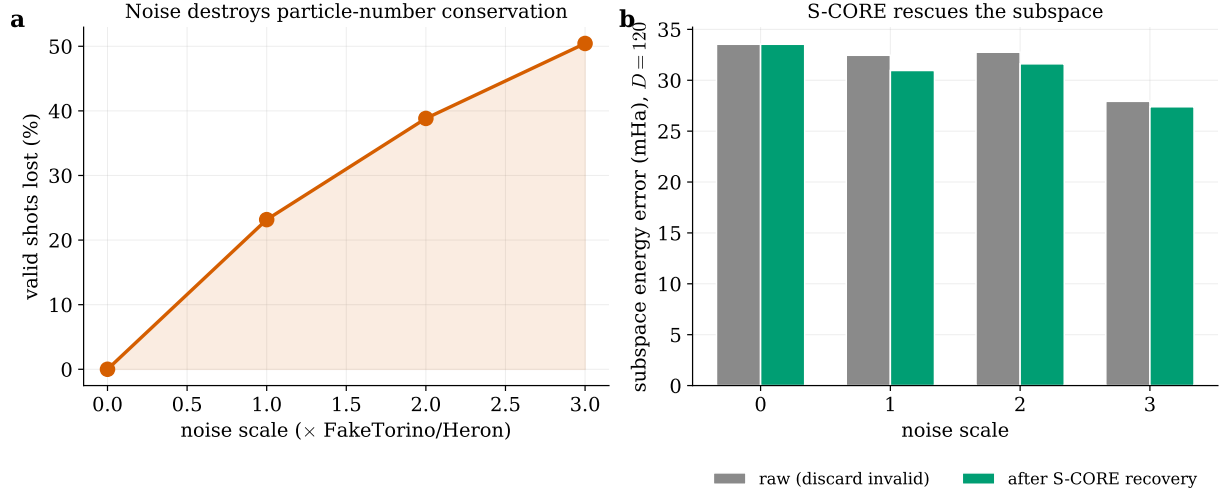


Figure 4. Configuration recovery under backend-calibrated noise (FakeTorino / Heron r1 rates; asymmetric readout + depolarizing). *Left:* the fraction of particle-number-valid shots collapses as the noise scale grows. *Right:* S-CORE recovery restores the subspace energy at fixed dimension $D = 120$, turning otherwise-discarded broken samples back into valid configurations.

the fraction of particle-number-valid shots collapses, yet S-CORE recovery restores the subspace energy at fixed dimension by imputing the broken samples back onto valid configurations.

3.3 Why the standard error-mitigation toolkit does not transfer

A methodological subtlety that the field has had to internalize, and that a review should make explicit, is that SQD is a *per-sample* post-selection pipeline, not an expectation-value pipeline. This distinction determines which of the near-term error-mitigation techniques transfer. Post-selection on particle number and spin, self-consistent recovery, and symmetry projection via classical shadows [Huang et al. 2020] operate on individual bitstrings and are central to the method. In contrast, the techniques that dominate IBM’s utility-scale expectation-value experiments — twirled readout error extinction (TREX), zero-noise extrapolation (ZNE), and probabilistic error cancellation (PEC) — mitigate errors *in expectation values*, not in individual samples, and therefore do not transfer to a pipeline that needs valid configurations rather than corrected averages. Dynamical decoupling

and Pauli twirling give only marginal, inconsistent benefit once the recovery loop has converged [Bhuiyan et al. 2026]. The practical consequence, relevant to §6, is that a claim about SQD’s hardware performance must be benchmarked with the mitigation stack it actually uses (recovery + symmetry post-selection), not the expectation-value stack, and that the shot budget saturates surprisingly early — optimal at 10^3 – 10^4 shots, with 10^5 performing *worse* [Bhuiyan et al. 2026]. This early hardware saturation does not contradict the $\sim 10^{14}$ -sample extrapolation of §3.1: that figure is the noiseless discovery limit for the exact heavy tail, whereas this one is the point at which, on noisy hardware with recovery already converged, additional shots mostly re-collect configurations the recovery step has restored rather than discovering new tail determinants.

4. Machine learning and generative methods for determinant and subspace selection

The proposition uniting this section is simple to state: if the important determinants are heavy-tailed and expensive to discover by sampling (§3.1), replace or augment the raw sampler with a *model* that has learned where they are. The proposition is also older than SQD — Coe’s machine-learning configuration interaction (MLCI) trained a neural network on-the-fly to predict which unseen determinants a classical selected-CI loop should add, on CO and stretched water, as early as 2018 [Coe 2018; 2019]. What SQD added is a reason to revisit the idea urgently and at scale, and a quantum sampler to compete with or complement. The result is a fast-moving ecosystem that, to our knowledge, has not been surveyed. We organize it along two axes — the *object* a method generates (a determinant, a subspace, a circuit) and the *importance signal* it is trained on (CI-coefficient magnitude, a perturbative estimate, quantum-sample frequency, or the diagonalized energy itself) — and we treat the four families in turn, closing with the conspicuous gap the taxonomy exposes.

4.1 A taxonomy by generated object and importance signal

Table 2 and Fig. 5 arrange the methods on the two axes. Two observations organize the rest of the section. First, the field is converging, from several directions, on a single algorithmic skeleton — *propose configurations, diagonalize in their span, use the result to improve the proposer, iterate*. This is not a coincidence but an attractor: the variational guarantee (§2.1) makes any proposal cheap to *score* by exact diagonalization, and a cheap exact score is precisely what turns configuration selection into a learning problem, pulling classical and quantum approaches alike toward the same loop. The consequence reframes the field — the SQD loop is *proposer-agnostic*, the quantum sampler one proposer among many (heat-bath screening, an RBM, an autoregressive network, a diffusion model, a GFlowNet), and the scientific question reduces to *which proposer wins in which regime*. Read this way, the GFlowNet whitespace of §4.6 is not one more idea but the one systematic gap in the space of proposers the attractor admits. Second, almost every method to date is a *distribution-matching* or *classification* model (it learns to reproduce, or to label as important, the determinants it has already seen); none, before the gap discussed in §4.6, is a *reward-proportional amortized sampler* explicitly built to place mass on diverse, high-importance determinants it has *not* yet seen — which is precisely the tail-discovery problem the coupon-collector law makes hard.

4.2 Restricted Boltzmann machines for generative recovery

The most direct instantiation of “learn the distribution of important determinants and resample it” uses a restricted Boltzmann machine (RBM). The Maitra group at IIT Bombay is the most active locus here. Their PIGen-SQD couples a physics-informed generative model to the SQD recovery step, using implicit low-rank tensor decompositions to steer the generator into the dominant sector, and — uniquely among the ML-for-SQD works — validates on real IBM Heron hardware [Patra et al. 2025]. Its density-matrix-embedding successor, QSCI-RBM, learns the distribution of dominant determinants from quantum samples and generates high-probability configurations,

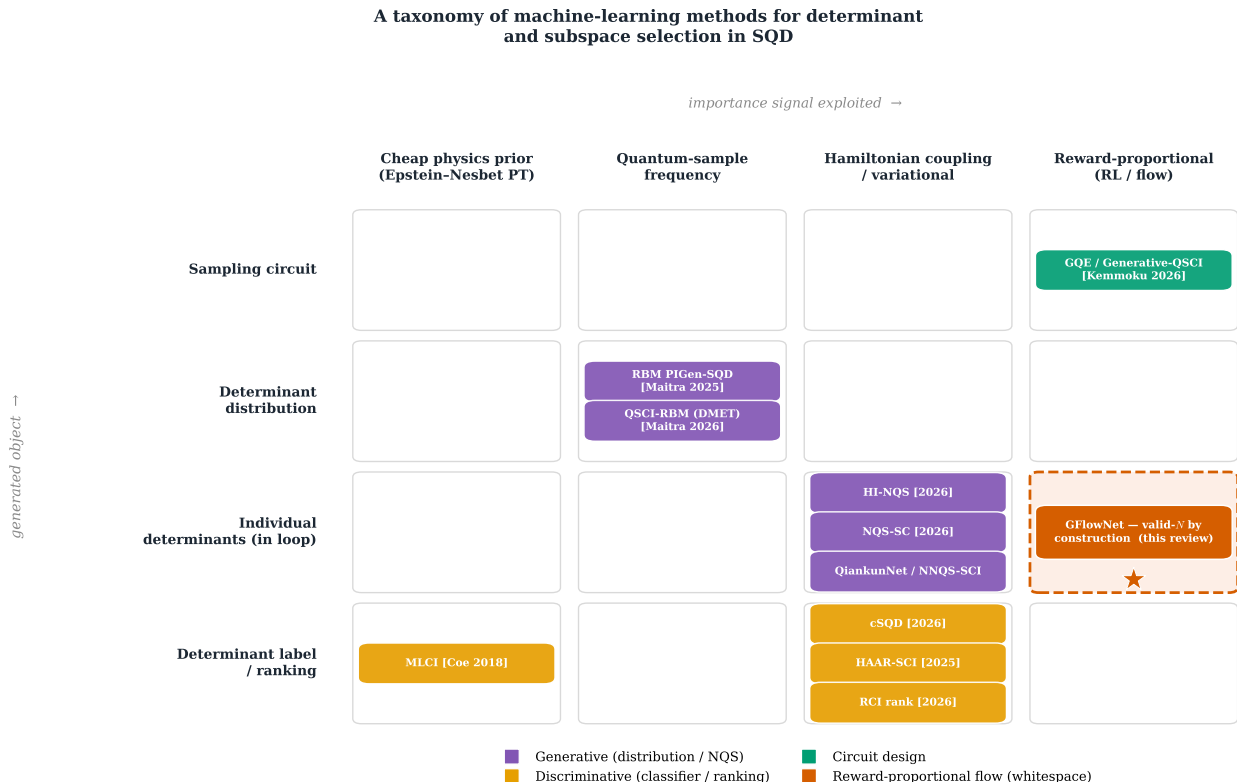


Figure 5. A taxonomy of machine-learning methods for determinant and subspace selection in SQD, organized by the object each method generates (rows) and the importance signal it exploits (columns). Generative models (purple) learn a distribution over determinants; discriminative models (orange) classify or rank them; circuit-design methods (green) learn the sampler itself. The dashed cell marks the methodological whitespace this review identifies: a reward-proportional generative-flow network that constructs valid- N determinants in the loop — unoccupied as of mid-2026.

reaching chemical accuracy on a SARS-CoV-2 main-protease-inhibitor complex while accessing only $\sim 4\%$ of the configuration subspace, against $\sim 20\%$ for standard SQD [Patra et al. 2026]. An independent lineage, building on generative-ML-in-configuration-space [Herzog et al. 2022], uses an interpretable RBM to solve the CI problem directly, recovering 99.99% of the correlation energy with orders of magnitude fewer determinants [Hernandez-Martinez et al. 2025]. The strength of the RBM route is that it is a genuine generative model of the determinant distribution; its limitations, relevant to §4.6, are that Gibbs sampling is iterative and mode-seeking, and that particle-number and spin symmetry are enforced by post-hoc filtering rather than by construction. (We note, for completeness and because the review record should be accurate, that an early QSCI-RBM preprint was withdrawn by arXiv administrators for a licensing-rights reason — not a scientific retraction — and its content re-appeared, expanded, in the density-matrix-embedding paper above.)

4.3 Autoregressive and transformer neural quantum states in the loop

A second, and currently the most vigorous, family replaces the quantum sampler with an autoregressive or transformer *neural quantum state* (NQS) [Carleo–Troyer 2017] that generates occupation-number bitstrings by construction with fixed particle number and exact ancestral sampling. The striking development of 2025–2026 is that this classical community has converged, independently, onto the SQD paradigm. HI-NQS embeds a dual-channel transformer — with explicit spin-up/spin-down cross-attention encoding fermionic structure — inside a sample-diagonalize-distil loop that its authors explicitly call sample-based quantum diagonalization, driven by a neural network rather than a circuit, and reports roughly half the determinant count of CIPSI on an N_2 active-space series [Chang et al. 2026]; a parallel autoregressive-NQS selected-CI, ARNN-

Table 2. Machine-learning and generative methods for determinant/subspace selection, organized by the object generated and the importance signal exploited. HW? indicates whether a quantum-hardware demonstration was reported.

Method	Generated object	Importance signal	HW?	Ref.
MLCI	determinant (add/skip)	on-the-fly NN prediction	classical	[Coe 2018]
PIGen-SQD	determinants (recovery)	quantum-sample freq. + physics prior	IBM Heron	[Patra et al. 2025]
QSCI-RBM (DMET)	determinant distribution	learned Born distribution (RBM)	classical	[Patra et al. 2026]
HI-NQS	determinants (in loop)	PT score + distilled eigenvector	classical GPU	[Chang et al. 2026]
NQS-SC	selected-config energy	variational, on selected set	classical	[Solanki–Ding–Reiher 2026]
NNQS-SCI / QiankunNet	determinants	autoregressive Born + de-dup	classical HPC	[Shang et al. 2025; Sun et al. 2026]
cSQD	determinant label	binary classifier + active learning	classical (+SQD)	[Zeni et al. 2026]
HAAR-SCI	determinants	Hamiltonian coupling (gated transformer)	classical GPU	[Zhang et al. 2025]
RCI (learning-to-rank)	determinant ranking	pairwise rank (transformer)	classical	[Nie et al. 2026]
Generative-QSCI (GQE)	circuit (not determinants)	QSCI subspace energy (RL policy)	classical sim.	[Kemmoku et al. 2026]
GFlowNet (proposed)	determinant, valid-N by construction	reward $\propto$ fused EN + sample freq.	— (whitespace)	<i>this work, §4.6</i>

SCI, uses the network to guide subspace expansion directly [Thompson–Gunlycke 2026]. Solanki, Ding, and Reiher’s NQS-SC evaluates the energy directly on a selected-configuration set and argues, on grounds of systematic improvability, that it should replace variational NQS as the default for electronic structure [Solanki et al. 2026]. At the high-performance-computing end, the QiankunNet line scales a transformer NQS-driven selected-CI engine to Hilbert spaces beyond 10^{14} , with GPU-accelerated de-duplication and coupled-configuration kernels [Shang et al. 2025; Sun et al. 2026]; a transformer-refined QSCI in this line has now been executed on the Zuchongzhi 3.1 superconducting processor, reaching chemical accuracy on a nitrogenase P-cluster model [Zeng et al. 2026]. These methods are, in effect, a classical mirror of SQD: the same loop, the same subspace diagonalization, but with a trained network as the proposer. Their relevance to a review of *quantum*-sampled diagonalization is twofold — they are the most natural classical baseline against which any quantum-sampling advantage must be measured (§5), and their architectural choices (symmetry-by-construction, distillation of the diagonalized eigenvector back into the proposer) are the design vocabulary any quantum-integrated learned proposer will draw on.

4.4 Neural-network classifiers and learning-to-rank selectors

A third family declines to model the full distribution and instead learns only to *rank* or *classify* determinants by importance — a lighter, and in practice highly effective, target. Zeni and co-workers recast determinant selection as binary classification inside an active-learning loop, with a classical variant (cHCI) cutting per-iteration memory roughly fivefold and a quantum–classical variant (cSQD) converging in markedly fewer SQD iterations [Zeni et al. 2026]. HAAR-SCI samples determinants autoregressively with a gated transformer and retains only those with the largest

Hamiltonian couplings via GPU min-heap kernels, reaching a mean absolute error of 0.51 mHa across eighteen molecules and, notably, beating semistochastic heat-bath CI on an iron–sulfur cluster while retaining under 0.01% of the Hilbert space [Zhang et al. 2025]. A learning-to-rank formulation with a transformer reaches chemical accuracy on iron–sulfur clusters using 12% of the CI space and reports beating SHCI on the notoriously multireference Cr_2 [Nie et al. 2026]; and a natural-orbital-based neural-network CI selector chooses determinants using approximate natural orbitals drawn from intermediate many-body eigenstates rather than Hartree–Fock orbitals [Thirion et al. 2025]. These results matter for the honest assessment of §5 in a specific way: several of them beat *classical* selected-CI — but they do so with a *classical* learned model, using no quantum hardware, which sharpens rather than resolves the question of what the quantum sampler adds.

4.5 Generative circuit design

A fourth family is generative in a different sense: rather than generate determinants, it generates the *circuit* that produces them. The generative quantum eigensolver (GQE) trains a transformer policy — a GPT-style model over circuit tokens — to design ansätze with desired properties, sidestepping the barren-plateau pathology of gradient-based variational optimization [Nakaji et al. 2024]. Applied to QSCI, this becomes generative circuit design in which the transformer is trained on the QSCI subspace energy to produce the state-preparation circuit whose samples define the subspace; on N_2 up to 32 qubits it reaches chemical accuracy with 98% fewer two-qubit gates than first-order Trotterization and yields subspaces roughly half the size of heat-bath CI in strongly correlated regimes [Kemmu et al. 2026]. This is the closest existing work to “a generative model that produces the SQD sampler,” and — a point we return to in §4.6 — it deliberately uses a transformer policy trained by reinforcement-style objectives, not a reward-proportional generative sampler.

4.6 The generative-flow-network gap, and the transferable toolbox

Surveying the four families above against the machine-learning literature exposes a conspicuous and, we argue, consequential gap. Generative flow networks (GFlowNets) [Bengio et al. 2021; Malkin et al. 2022] are amortized samplers purpose-built to generate compositional discrete objects with terminal probability proportional to a reward, favouring *diverse* high-reward states rather than collapsing onto the mode — a property established both empirically and through their equivalence, in expected gradient, to a hierarchical variational inference that avoids the mode-seeking of reverse-KL objectives [Malkin et al. 2023]. They have been applied across molecular graphs, crystals, biological sequences, combinatorial optimization, and Bayesian structure learning, and even to adjacent quantum tasks — variational-circuit design, unitary synthesis, and measurement grouping (see [Bengio et al. 2021] and references therein). Yet an exhaustive search across every relevant term (GFlowNet with configuration interaction, determinant, selected-CI, Slater determinant, SQD, QSCI, Fock space) returns, as of mid-2026, *no* application of a GFlowNet to determinant or subspace selection for configuration interaction. The nearest work is the generative-circuit QSCI of §4.5, which uses a transformer policy and deliberately not a GFlowNet. The gap is not an accident of terminology but a genuine methodological opening: a determinant is a fixed-particle-number occupation bitstring built one orbital at a time — precisely the compositional discrete object a GFlowNet constructs — and the correlation energy lives in the *many small-weight determinants of the tail*, the regime where a reward-proportional sampler that resists mode collapse is exactly the right tool and a reward-maximizing search is exactly the wrong one. We are careful about the scope of the resulting novelty claim. That a *generative* model helps SQD is not novel (§4.2–§4.5); that noise can help recovery is not novel (§3.2). What is unclaimed is the specific combination — a GFlowNet whose forward construction of determinants is masked to the fixed-particle-number sector by construction, trained to sample proportional to a fused importance reward — together with the controlled, exact-ground-truth study (§6) that would establish *when* such a proposer helps. The adjacent literature already supplies the toolbox for it: energy-based GFlowNets as amortized samplers of unnormalized

discrete distributions [Zhang et al. 2022], “inexpensive-reward” pretraining that mirrors the cheap-perturbative-estimate-then-expensive-diagonalization structure of the problem [Pandey et al. 2024], temperature-conditional reward tempering that is a learnable analogue of CIPSI’s threshold schedule [Kim et al. 2024], and discrete-diffusion set-selection models for which a determinant subspace is literally a binary selection mask [Sun et al. 2023]. Whether any of these transfers *usefully* — and whether it survives the classical baselines of §5 — is, we argue, the central open experimental question of the subfield.

Concretely, such a proposer needs a reward that is cheap and FCI-free. The natural choice — the same criterion CIPSI and heat-bath CI use to grow their subspaces — is the Epstein–Nesbet first-order coefficient of each determinant relative to the Hartree–Fock reference,

$$c_i^{(1)} = \frac{\langle D_i | \hat{H} | D_{\text{HF}} \rangle}{E_{\text{HF}} - \langle D_i | \hat{H} | D_i \rangle}, \quad R(i) \propto (c_i^{(1)})^2, \quad (4)$$

computable with a single $\hat{H}|D_{\text{HF}}\rangle$ matrix–vector product and the Hamiltonian diagonal. A GFlowNet with forward policy $P_F(a_t | s_{t-1}; \theta)$ builds a determinant as a trajectory $\tau : s_0 \rightarrow s_1 \rightarrow \dots \rightarrow x$ that places electrons one orbital at a time — masked so that x is always a valid n_α -electron string — and is trained by the trajectory-balance objective

$$\mathcal{L}_{\text{TB}}(\tau) = \left(\log Z_\theta + \sum_t \log P_F(a_t | s_{t-1}; \theta) - \log R(x) \right)^2, \quad (5)$$

whose global minimizer samples terminal states in exact proportion to reward, $P_\theta^\top(x) \propto R(x)$ — the diverse, tail-covering behaviour the coupon-collector regime demands. Fig. 6 evaluates exactly this construction on the FCI-verifiable N_2 system. Given only the cheap, FCI-free reward, the GFlowNet learns to sample the important region and tracks the (unreachable) exact oracle far better than blind uniform sampling; naive i.i.d. importance sampling from the same peaked reward, by contrast, collapses onto a handful of modes and saturates below dimension 30 — the coupon-collector wall in miniature, and a first concrete illustration of the diverse-sampling behaviour a GFlowNet is built to provide. This comparison isolates the sampling law, not the architecture: we do not yet benchmark against autoregressive neural quantum states with exact ancestral sampling — the strongest learned proposer, §4.7. The decisive comparison, however, is not against a static top- K ranking but against the strong *iterative* heat-bath CI that re-bootstraps its selection from the current correlated wavefunction; that comparison, deferred to Fig. 7 in §5, is where the classical selector wins at every FCI-verifiable scale. The honest noise-free verdict is therefore that a learned proposer, however well it tracks the oracle, does not beat strong classical selected CI — and that this changes only under noise, the subject of §6–§7.

4.7 A note on real-space neural wavefunctions

For completeness and to forestall a common conflation, we distinguish the methods above from the real-space neural-wavefunction programme — FermiNet, PauliNet, Psiformer, and the emerging foundation models Neural Pfaffians and Orbformer [Pfau et al. 2020; Hermann et al. 2020; von Glehn et al. 2023; Gao–Günemann 2024; Foster et al. 2025]. These are variational Monte Carlo ansätze in continuous three-dimensional space, not determinant selectors in a fixed orbital basis; they are among the strongest classical methods for small strongly correlated molecules and thus appear in §5 as competitors — the discrete-orbital neural-network backflow of Liu and Clark reaches state-of-the-art molecular energies among them [Liu–Clark 2024], whose optimized variant now matches heat-bath CI, ASCI, FCIQMC and DMRG and surpasses coupled cluster — making it the strongest single learned proposer to date [Liu–Clark 2025], and the programme is surveyed by [Hermann et al. 2023] — but they do not propose configurations for a diagonalization and are outside the loop this review concerns. The one bridge worth flagging is the discrete-basis autoregressive NQS

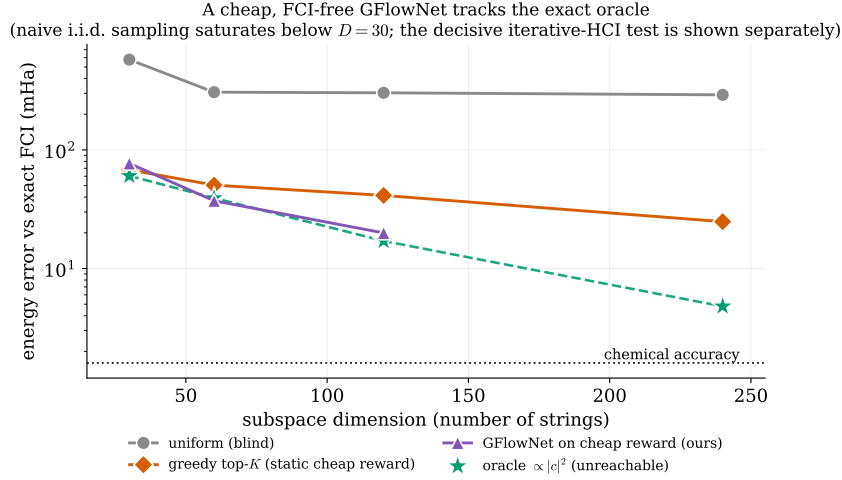


Figure 6. Compactness under a cheap, FCI-free reward (N_2 , exact FCI). Energy error vs subspace dimension for blind uniform sampling, deterministic top- K selection by the static cheap Epstein–Nesbet reward, the GFlowNet trained on the same (tempered) cheap reward, and the unreachable exact oracle. The GFlowNet learns the important region and tracks the oracle far better than blind sampling; naive i.i.d. sampling from the peaked reward saturates below $D=30$ and is off-scale. The GFlowNet curve terminates at $D=120$ because its tempered ($\beta=0.5$) sampler concentrates mass on the ~ 140 highest-reward determinants — matching the exact oracle there (≈ 17.7 vs 17.0 mHa) while blind uniform sampling stalls near 240 mHa — so covering a larger subspace would call for the temperature schedule that is §4.6’s learnable analogue of the CIPSI threshold. The decisive comparison against the stronger *iterative* heat-bath CI is Fig. 7, where the classical selector wins at every FCI-verifiable scale.

(NAQS) [Barrett et al. 2022], which *does* sample determinants proportional to $|\psi|^2$ and is therefore the closest classical relative — and the most natural benchmark — for any learned determinant proposer, GFlowNet-based or otherwise.

5. The critical assessment: does the quantum sampler, or its ML augmentation, beat classical selected CI?

Every method in §4 is ultimately in service of an SQD subspace, and every SQD subspace inherits a single, unavoidable question. Because the final step is an exact diagonalization in a determinant subspace, SQD *is* a selected configuration interaction whose determinants happen to be chosen by a quantum sampler. Its natural competitors are therefore the classical selected-CI heuristics — and those are neither weak nor slow. A review that omitted this comparison would be advocacy, not assessment; we make it the centre of the paper.

5.1 The classical baselines and the accuracy bar

The relevant classical methods fall into two groups. The direct competitors are the classical determinant selectors: CIPSI, with its Epstein–Nesbet perturbative selection and extrapolation to the FCI limit [Huron et al. 1973; Garniron et al. 2019; Loos et al. 2024]; heat-bath CI (HCI) and its semistochastic form (SHCI), whose $|H_{ai}c_i|$ screening makes selection near-linear and which is generally the fastest route to sub-milli-hartree accuracy [Holmes et al. 2016; Sharma et al. 2017]; and adaptive sampling CI (ASCI) [Tubman et al. 2016]. The best-in-class solvers for strong correlation are DMRG [White 1992; Chan–Head-Gordon 2002], phaseless AFQMC [Zhang–Krakauer 2003], full configuration-interaction quantum Monte Carlo [Booth et al. 2009], and, for weak-to-moderate correlation, coupled cluster. Each of these baselines has a canonical, openly available implementation against which any claim can be reproduced — Dice for (semistochastic) heat-bath CI, Quantum Package for CIPSI, block2 for DMRG, ipie for AFQMC, and NECI for FCIQMC, all interoperable

through PySCF — and we take it as part of the standard below that a benchmark name the baseline code it was run against. The accuracy these methods agree on sets the bar any quantum result must clear to matter, and that bar is now quantitative: in the blind benzene ground-state challenge — a (30e,108o) space across seventeen groups — DMRG, AFQMC, FCIQMC, and SHCI concur on the correlation energy to a root-mean-square deviation of 1.3 mHa [Eriksen et al. 2020], and the Simons hydrogen-chain benchmark reaches comparable cross-method concordance in the thermodynamic limit [Motta et al. 2017]. The same collaboration’s transition-metal capstone — twenty first-principles methods on 3d atoms, ions, and monoxides, with experiment-free reference values [Williams et al. 2020] — fixes the classical bar in exactly the strongly-correlated, 3d/iron-sulfur arena that SQD most wants to claim, and a curated open hierarchy of chemically decisive multireference systems (N_2 , FeS , $[\text{2Fe-2S}]$, U_2) has since been assembled to standardize such comparisons [Sundar et al. 2026]. The operative standard, then, is not “reaches chemical accuracy” but “reaches the ~ 1.3 mHa cross-method consensus at *lower cost* than these methods, or in a regime where they all degrade.” No SQD result has yet been measured against this bar and passed; indeed the largest systematic hardware benchmark to date — 745 reactions on a superconducting processor — finds that SQD attains CCSD-level accuracy only after energy extrapolation, not directly [Raisuddin et al. 2025].

5.2 The critique literature

Two peer-reviewed-track critiques frame the negative case. Reinholdt and co-workers, granting SQD an idealized noiseless sampler, show on N_2 and $[\text{2Fe-2S}]$ — the very systems used to showcase the method — that its expansions are roughly an order of magnitude less compact than heat-bath CI (for $[\text{2Fe-2S}]$, 50,551 determinants against HCI’s 4,260 at 0.1 Ha error), and that the coupon-collector cost to reach micro-hartree precision extrapolates to $\sim 10^{14}$ samples; their conclusion is that QSCI “falls behind more effective classical selected-CI heuristics” [Reinholdt et al. 2025]. Gaberle and Jattana sharpen the claim for strongly correlated lattice models: the number of configurations required for a fixed accuracy grows *exponentially with system size even under optimal, decreasing-probability inclusion* — that is, even a *perfect* sampler fails, so the limitation is intrinsic wavefunction delocalization, not a sampling artefact [Gaberle–Jattana 2026]. This second result is the more fundamental, and to date it has no proposed remedy.

5.3 The 2026 dequantization

The critique became concrete in 2026. Belagali and co-workers gave a polynomial-time classical algorithm for the energy of any single-layer local unitary cluster Jastrow circuit — the exact ansatz SQD samples — exploiting its near-Gaussian (matchgate) structure, and reproduced the flagship 77-qubit $[\text{4Fe-4S}]$ experiment on a laptop in under a minute, finding, through classical circuit optimization, an energy *below* the hardware result [Belagali et al. 2026]. The authors are careful that this is weak (energy) rather than strong (bitstring) simulation, leaving a formal gap; but as a statement about the specific published experiments it is decisive. Independently, a classical iterative refinement seeded from random determinants (TrimCI) reached the $[\text{4Fe-4S}]$ ground state with on the order of 10^6 -fold fewer determinants than the quantum run [Zhang–Otten 2025], and the noise-and-recovery study of §3.2 showed random determinants plus classical recovery reproducing the accuracy [Vaquero-Sabater et al. 2026]. Taken together, these results locate the accuracy not in the quantum sampler but in the classical recovery-and-diagonalization back end. The trend has since consolidated. A fermionic-linear-optical (matchgate) simulation not only reproduces the LUCJ circuit classically but *improves* on the noisy-hardware SQD energy [Hassman et al. 2025] — unsurprising given device noise, the substantive point being that no quantum resource was needed, not that a new accuracy record was set. And, most tellingly, a bespoke classical eigenwalk solver from the same IBM authors who reported the 49-qubit advantage reportedly solves a subset of the instances underlying that claim classically [Chin et al. 2026], so the advantage question is now partly

self-refereed within a single group. In parallel, classical coupled-cluster and DMRG reached chemical accuracy on the FeMo-cofactor model long held up as the canonical quantum target [Zhai et al. 2026], and transformer neural-network backflow surpassed DMRG on [2Fe-2S], the very iron-sulfur system SQD showcases [Ma et al. 2025].

5.4 The pro-quantum case, weighed honestly

The defence is real but narrow. Hafid and co-workers prove that UCJ circuits can encode arbitrary instantaneous-quantum-polynomial (IQP) computations, so *some* such circuits are classically hard to sample unless the polynomial hierarchy collapses [Hafid et al. 2025] — a worst-case statement that does not describe the few-layer, chemically-structured LUCJ circuits actually run, which §5.3 shows are easy. Weaving and co-workers report time-evolved QSCI subspaces more than two hundred times more compact than a naive selected-CI criterion and “comparable to HCI,” but explicitly decline to claim a quantum advantage [Weaving et al. 2025]. The strongest experimental claim is Kirby and co-workers’ 49-qubit sample-based Krylov result, reported to outperform classical sparse-ground-state solvers [Kirby et al. 2026] — but on *sparse-ground-state local Hamiltonians*, not general chemistry, a scope that must be stated precisely and whose classical rebuttals should be watched. The one genuinely open crack is depth: the dequantization proofs are for single-layer LUCJ, and whether deep, multi-layer LUCJ evades classical simulation is unresolved. It is the only place a future chemistry advantage could currently hide.

5.5 Verdict

The honest state of the field, as of mid-2026, is that there is *no reproducible, same-active-space demonstration that the SQD/QSCI quantum sampler beats SHCI, CIPSI, DMRG, or AFQMC — or even random determinants passed through classical recovery — on molecular electronic structure*, and that the flagship sampling circuits have been classically reproduced at the energy (weak-simulation) level. This is not a claim that SQD is useless: it is a genuinely useful, noise-tolerant, well-tooled workflow that has reached chemically relevant scale, and its configuration-recovery idea is a real contribution. It is a claim that the burden of proof for *advantage* rests squarely on the quantum side, that this burden has not been met, and — the point of §6 and §7 — that meeting it, or honestly ruling it out, requires a standard of benchmarking the field has not yet adopted. This verdict rests on the independent critique and dequantization literature above — not on any single computation of ours — and now coincides with independent macro-assessments of quantum advantage in electronic structure [Lee et al. 2023; Gundlach et al. 2025; Legeza et al. 2026]. Fig. 7 merely illustrates it on two FCI-verifiable systems, where iterative heat-bath CI bootstrapped from the correlated wavefunction reaches 0.6 mHa on H₂O and ≈ 10 mHa on stretched N₂, at or below a noise-level quantum-sampled subspace.

6. A standard for honest benchmarking

The pattern that recurs across §5 is not that quantum methods were shown to fail, but that they were often not measured against the baselines that would settle the question. We therefore propose, and argue the field should adopt, a benchmarking standard with ten elements; we state it as a checklist because its value is in being applied uniformly. Systematic hardware benchmarking of SQD has recently begun [Raisuddin et al. 2025]; our emphasis differs in insisting on exact-FCI ground truth and an explicit dequantization baseline, and the checklist is exactly the protocol any defensible study of a learned proposer — ours or others’ — would follow.

(i) **Exact ground truth at verifiable scale.** Claims about which determinants matter, and how compact a subspace is, are only meaningful where the exact answer is known. The controlled experiment should live in active spaces small enough for exact FCI (up to roughly a dozen orbitals)

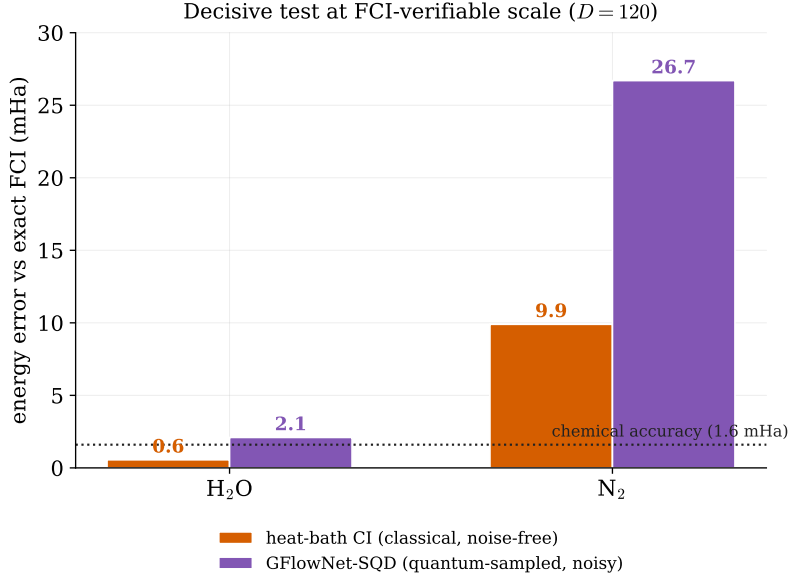


Figure 7. The decisive classical test at FCI-verifiable scale ($D = 120$). Iterative heat-bath CI, bootstrapped from the correlated wavefunction, reaches 0.6 mHa on H₂O and ≈ 10 mHa on stretched N₂ — at or below the noisy quantum-sampled subspace on every system. At verifiable scale, classical selected CI wins.

— large enough for the coupon-collector effect to appear, small enough that every energy is an error against truth, not against another approximation. This is a deliberate inversion of the “largest active space” instinct: the point is not scale but falsifiability.

(ii) The strong classical baseline, not a straw man. The comparison must be against iterative heat-bath / semistochastic CI bootstrapped from the correlated wavefunction — and, following §5.3, against the two baselines the field has repeatedly found decisive: random determinants passed through the classical recovery loop, and the classical simulation of the sampling circuit itself. A method that beats uniform sampling but not SHCI has demonstrated nothing about quantum value.

(iii) A dequantization baseline. Given [Belagali et al. 2026], any hardware claim should report whether the sampling circuit is classically simulable at the depth used, and if so, the energy that classical simulation attains. This is now as basic a control as reporting the classical FCI reference, and it has four independent routes the strongest studies report against wherever they apply: fermionic-linear-optical (matchgate) simulation [Belagali et al. 2026; Hassman et al. 2025], sparse guided eigenwalks [Chin et al. 2026], Pauli propagation [Shrikhande et al. 2025], and direct tensor-network (matrix-product-state) contraction of the circuit — the last two being the depth-truncable routes that most directly probe the multi-layer-LUCJ crack of §5.4.

(iv) Full cost accounting in the currency that is scarce. The scarce resource is quantum measurement (shots = QPU time), and the honest axis is energy error versus shots, or versus subspace dimension at matched classical diagonalization cost — not energy error versus wall-clock on incomparable hardware. Reporting the shot budget honestly also surfaces the early saturation of §3.3.

(v) Statistical error bars. Sampling is stochastic and neural proposers are seed-dependent; a single run is an anecdote. Multi-seed means and standard deviations, and a significance statement on any claimed separation, should be mandatory. In our own controlled work we found that a single-run “win” of a generative proposer over a classical control evaporated to a statistical tie once five-seed error bars were computed — a cautionary instance of exactly this point.

(vi) Backend-calibrated, asymmetric noise. Because SQD’s noise sensitivity runs through particle-number-violating readout errors (§3.2), a credible noise model must be backend-calibrated

and must include the T_1 -driven asymmetry of readout, which symmetric bit-flip models understate; and the analysis should include an ablation isolating that asymmetry. A readout-only model will, in turn, be criticized where gate errors dominate, so the strongest studies will move toward full noisy-circuit simulation or a confirmatory hardware point.

(vii) Convergence of the reduction to the model problem. Every SQD number is a claim about a chosen active space in a chosen basis; the benchmark should report active-space, basis-set, and orbital-choice (natural- versus canonical-orbital) convergence, so that an apparent advantage is not an artefact of a fortunate truncation.

(viii) Energy differences and properties, not only total energies. Chemistry is decided by relative quantities — barriers, spin gaps, dissociation energies — and a method can reach a good total energy while misranking the differences that matter; a benchmark should report at least one chemically decisive energy difference, not a single total energy.

(ix) More than one geometry. A single-point win can be a coincidence of the reference; several geometries along a dissociation or reaction coordinate test whether the advantage is systematic, and are the natural setting for the amortization question of §7.

(x) Pre-registered, commensurate total-resource accounting. Beyond the shot axis of (iv), the honest total cost is quantum and classical together — QPU-seconds reported alongside the classical node-hours of the diagonalization and recovery, in the commensurate-resource spirit of [Raisuddin et al. 2025] — and the seed count and significance threshold of (v) should be fixed before the runs, not chosen after.

We contend that a study meeting all ten is publishable and valuable *whatever its result* — a rigorous negative result that maps where a generative or quantum proposer does and does not help is, in a field this contested and this prone to under-benchmarked demonstrations, more useful than another hopeful positive that a strong baseline would deflate.

7. Outlook: where generative and quantum methods retain a defensible advantage

The verdict of §5 is easy to read as a counsel of despair; we read it instead as a map. The reason classical selected CI wins the SQD contest, on the systems studied, is structural: SQD’s task is *classically verifiable*, and — empirically, on molecular systems — the same structure that lets one check a subspace (a projected Hamiltonian one can diagonalize) also makes a cheap classical *importance heuristic* available with which to construct it; the dequantization of §5.3 is the extreme statement of the same fact, that even the sampler is classically reproducible. (This is an observation about the available classical signal, not a general implication of verifiability — verifiability alone need not imply cheap constructibility, as the P-versus-NP gap makes plain.) There is little quantum edge to be had where such a cheap classical construction happens to exist. This is not a wall but a compass: it says to aim generative and quantum methods at the regimes where a strong classical baseline provably *cannot* be cheap. This converts the compass from a metaphor into a falsifiable prediction the standard of §6 can test: if the locus of advantage is exactly where the cheap importance signal misranks the truly important determinants, then any genuine generative or quantum advantage on molecular electronic structure should appear first, and only, where the *rank correlation between the cheap prior (e.g. the Epstein–Nesbet coefficient) and the exact CI weights collapses*. That rank correlation is directly measurable at FCI-verifiable scale — our own MVE finds it already imperfect (log-rank-correlation ≈ 0.55 on N_2) — and we propose it as the order parameter for where to look: a single scalar that should predict, system by system, whether a learned or quantum proposer can help before any hardware is run. A method that helps where the prior already ranks well would falsify the picture; a method that helps only as the correlation degrades would confirm it. Three such regimes are visible.

Noise-limited recovery and genuine multireference character. Within SQD itself, the

Where a quantum or generative advantage can, and cannot, live

The structure that makes SQD classically *verifiable* is the structure that makes it classically *constructible*. Advantage survives only where that equivalence breaks.

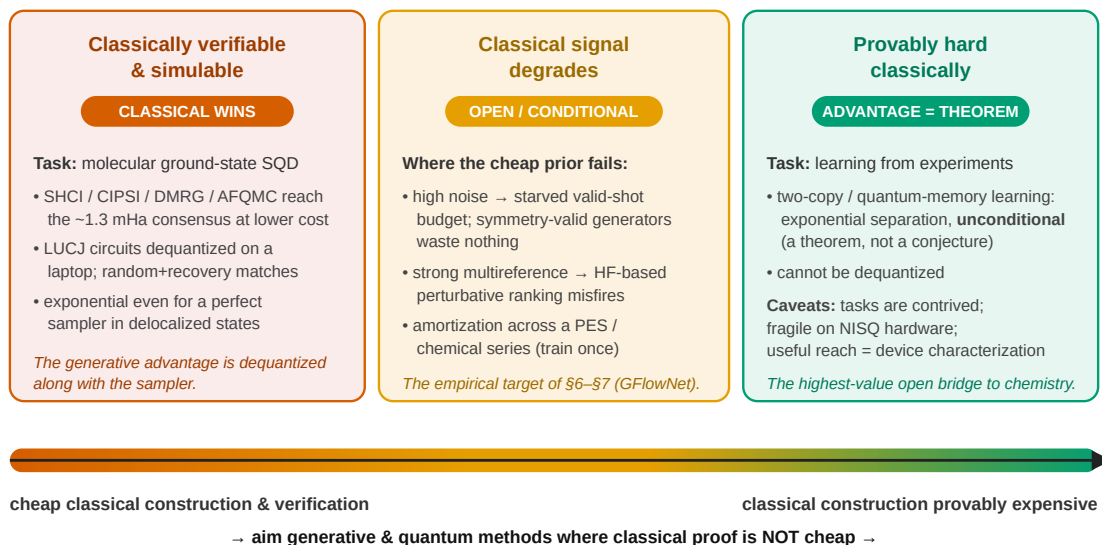


Figure 8. The advantage landscape of §5–§7. The structure that makes SQD classically *verifiable* is the structure that makes it classically *constructible*; advantage survives only where that equivalence breaks. Colour-blind-safe (Okabe–Ito) palette; zones are also distinguished by header text and verdict chips, not colour alone.

classical selectors win because a cheap classical importance signal (a Hartree–Fock-referenced perturbative estimate) is available and accurate. Where that signal degrades — in the high-noise regime, where the valid-shot budget is starved and a generator that emits only symmetry-valid configurations by construction wastes nothing, and in genuinely multireference systems, where a single-reference perturbative prior misranks the important determinants — a learned proposer that fuses the quantum-sample statistics with a cheap classical prior has room the *perturbatively-primed* classical selector does not. (Here the honest baseline is not only iterative SHCI but the classical learned rankers of §4.4, which already contest — and on Cr₂ reportedly beat — SHCI in exactly this multireference regime with no quantum resource; the live question is therefore the *incremental* value of the quantum samples fused into such a learned proposer, not of learning over perturbation theory per se.) Crucially, the fair classical control here is itself the noise-robust recovery loop of §5.3 (S-CORE plus the cheap prior that already reproduces the accuracy), so any generative gain is measured *beyond* that classical recovery — and, consistent with the statistical tie of §6(v), it is a small, conditional effect, not a reversal of the verdict. Establishing *whether* and *when* it uses that room is precisely the controlled, exact-ground-truth study the standard of §6 defines, and is the natural first empirical target for the GFlowNet route of §4.6. We emphasize that the honest expectation, from the trajectory of §5, is a *conditional* advantage — a regime map, not a universal win — and that a paper reporting such a map, negative regions included, is the valuable contribution. Fig. 9 is that map in miniature, computed on the FCI-verifiable N₂ system: the energy-error gap between the fair classical control and the fused generative proposer is *negative* at zero noise (the classical prior alone suffices) and turns increasingly positive as the device-noise scale grows and the valid-shot budget starves — the crossover that localizes the generative advantage to the noise-limited regime.

Amortization across chemical space. A learned proposer offers something a per-instance classical selector does not: amortization. A model conditioned on molecular geometry and trained once can, in principle, generate compact subspaces across an entire dissociation curve or chemical series without re-running an expensive selection at each point — the value proposition of the

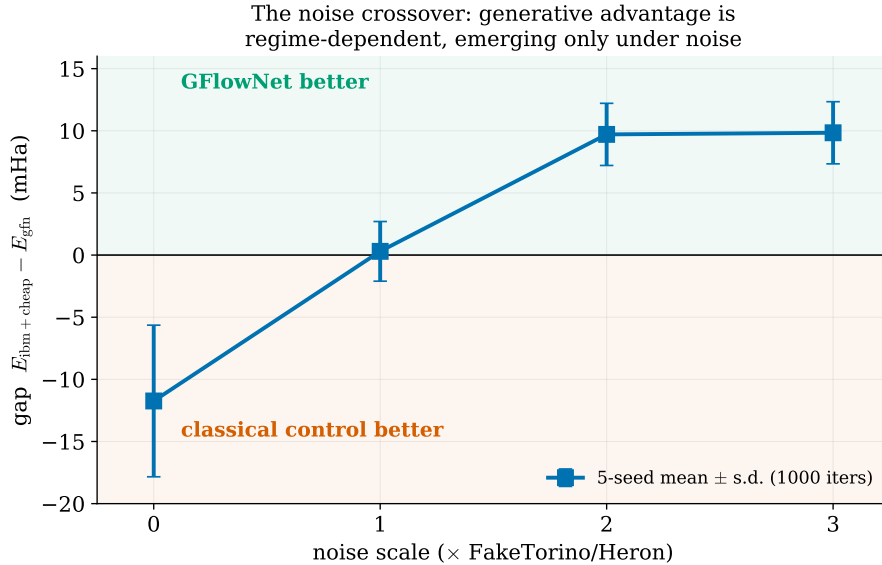


Figure 9. The noise crossover. Energy-error gap between the fair classical control (S-CORE recovery + cheap-prior fill) and the fused generative proposer, vs device-noise scale, over random seeds (mean $\pm$ s.d.). At low noise the classical control is better (negative gap); as noise grows the constraint-respecting generative proposer pulls ahead. The advantage is regime-dependent, not universal.

foundation-wavefunction programme [Foster et al. 2025], transposed to the determinant-selection setting. Here the metric is not accuracy at one geometry but total cost across many, and it is a genuinely open, and genuinely ML-flavoured, question whether amortization beats re-running SHCI per point.

Learning from experiments: where the advantage is a theorem. The deepest reason for optimism lies just outside the SQD loop, and it inverts the logic of §5 exactly. The one place in this landscape with an *unconditional, information-theoretic* quantum advantage is the quantum-data-native task of learning properties of an unknown quantum state or process: a learner with quantum memory that measures two coherent copies jointly needs exponentially fewer experiments than any single-copy strategy, a separation proved without complexity assumptions [Chen–Cotler–Huang–Li 2021] and demonstrated on superconducting and photonic hardware [Huang et al. 2022; Liu et al. 2025], and surveyed among the field’s few unconditional advantages by [Huang et al. 2025]. Unlike SQD, this advantage cannot be dequantized, because its classical lower bound is a theorem, not a conjecture about circuit structure. This suggests a reorganizing principle for the enterprise as a whole: candidate quantum advantages should be sorted not by their promised magnitude but by the *type* of their classical lower bound. Those resting on a complexity-theoretic conjecture — the presumed hardness of sampling a particular circuit family — are perpetually one clever classical algorithm away from collapse, as SQD’s own two-year arc from flagship experiment to laptop reproduction (§5.3) attests; those resting on an unconditional, information-theoretic separation are not, because no future algorithm can close a gap that is already a theorem. A field repeatedly surprised by dequantization would do well to spend its scarce hardware where the lower bound is a theorem, and to treat conjecture-backed advantages as demonstrations of scale, not of separation, until proven otherwise. Its honest caveats are equally sharp and belong in any forward-looking account: the demonstrated-advantage *tasks* (predicting Pauli expectations, purity, and random-channel characteristic functions) are information-theoretically natural but scientifically contrived; the advantage is fragile on superconducting hardware, degrading under the depolarizing noise of a heavy-hex device [Cotler–Gong–Kannan 2025] to the reduced, mitigation-propped separation IBM has actually measured [Seif et al. 2024]; and the one *useful* application delivered with rigorous backing is the characterization of quantum devices themselves. The bridge from this provable-but-contrived frontier to genuinely useful chemistry is, we argue, the highest-value open problem adjacent

to the subfield — and the natural destination for the generative machinery reviewed here. That bridge might be built by learning a device’s effective error model — as machine-learning quantum error mitigation already does at utility scale [Liao et al. 2024] — to improve an SQD estimator, or by learning the properties of a molecular state prepared on hardware.

8. Conclusions and open problems

Machine learning entered sample-based quantum diagonalization for a sound reason — the configuration-recovery step is a selection-and-denoising problem that a learned model should do better than an occupancy-weighted heuristic — and, in two years, has produced a rich ecosystem of restricted Boltzmann machines, neural-quantum-state proposers, classifiers, ranking models, and generative circuit designers. This review has organized that ecosystem, mapped its methods and open-source tooling, and — most importantly — insisted on holding it to the question it exists to answer. The honest answer, at present, is that neither the quantum sampler nor its machine-learning augmentation has been shown to beat classical selected configuration interaction on molecular electronic structure at any verifiable scale, and that the flagship sampling circuits have been classically reproduced at the energy (weak-simulation) level. We have argued that this is a compass rather than a verdict of futility: it points generative methods toward the noise-limited and multireference regimes where the cheap classical signal fails, toward amortization across chemical space, and, as an adjacent horizon, toward the quantum-data-native learning tasks where the advantage is provable but not yet connected to useful chemistry.

Four open problems structure the road ahead. First, the *intrinsic-delocalization* barrier of [Gaberle–Jattana 2026] — that some strongly correlated ground states require exponentially many configurations even under a perfect sampler — has no proposed remedy and may be fundamental. Second, the *depth* question: whether deep, multi-layer LUCJ circuits evade the classical simulability that [Belagali et al. 2026] proved for the single-layer case is the one place a chemistry advantage could still live, and settling it is urgent. We flag a structural worry worth confronting rather than postponing. The VQE programme collapsed when it emerged that the conditions letting a circuit avoid barren plateaus tend also to render it classically simulable [Cerezo et al. 2025]; SQD may face an isomorphic dichotomy in a different dialect, since the shallow, structured, low-entanglement circuits whose samples are noise-tolerant and amenable to configuration recovery are precisely those most exposed to matchgate-style dequantization [Belagali et al. 2026], while the depth that would defeat classical simulation may also be the depth at which readout noise and the coupon-collector tail (§3.1) overwhelm the loop. Whether a usable corridor exists between these two failure modes is, we conjecture, the question on which a near-term *chemistry* advantage for SQD ultimately rests — and it is one the field has the tools to settle. Third, the *useful-target* gap in learning-from-experiments — bridging provable-but-contrived tasks to science anyone needs — is the highest-value problem adjacent to the field. Fourth, and most immediately actionable, the *GFlowNet gap*: the one generative paradigm built for diverse, reward-proportional sampling over exponential discrete spaces has not been applied to determinant selection, and whether it helps — measured, per §6, against exact ground truth and strong classical baselines — is a clean, falsifiable, and currently unclaimed question. We have tried to write the review such that its most useful legacy would be not a catalogue but a standard: that the next claim in this field, positive or negative, arrives with exact ground truth, strong and dequantization baselines, honest cost accounting, and error bars — so that the field can, at last, know what it knows.

Limitations of this review. This survey is a snapshot of a subfield that is expanding on a timescale of weeks; a substantial fraction of the primary literature we cite is preprint and not yet peer-reviewed, and several attributions and identifiers should be re-verified against the published versions. Our coverage extends to mid-2026 and will inevitably miss work that appears between writing and publication — including, very possibly, the first application of a generative-flow network to determinant selection whose absence we here record. We have tried to make the review robust to that churn by organizing it around durable questions (compactness, verifiability, the locus of advantage)

and a benchmarking standard, rather than around any single result. Finally, a positional disclosure: the authors are actively developing the generative-flow-network proposer whose absence this review documents, and the standard of §6 is the protocol we intend that study to meet; we have therefore been deliberate in presenting the GFlowNet gap as an open, falsifiable question — one our own forthcoming controlled study may well answer in the negative — rather than a solved one, and in resting the §5 verdict on the independent literature rather than on our own illustrative figures.

Data and code availability

Every quantitative claim in this review is reproduced from executable code in the companion computational notebook `GFlowNet_SQD_calculations.ipynb`, which runs in Google Colab or any Python ≥ 3.10 with `pyscf`, `torch`, `scipy`, and `matplotlib`. The notebook regenerates, from first principles and with fixed random seeds, the exact FCI reference of Fig. 1’s system, the coupon-collector statistics of §3, the S-CORE recovery of §3.2, the Epstein–Nesbet reward and GFlowNet compactness study of §4, the noise-crossover of §6, and the heat-bath-CI decisive test of §5. The complete code — the notebook, the calculation scripts behind every reported number, one script per figure, the generated figures, and the Docker build environments — is openly available (MIT license) at <https://github.com/nicolasbonilla/ml-for-sqd-review>, with a one-click Google Colab entry point and every notebook cell and figure rendered inline in the browser. Figs. 1 and 2 were produced as vector graphics and are available with the source. No proprietary data or hardware access is required to reproduce any figure or table; all “real-noise” results use local, backend-calibrated noise models (`FakeTorino`, IBM Heron r1).

Author contributions

N.B.V. conceived the review, performed the literature synthesis, designed and ran all computational experiments, produced the figures, and wrote the manuscript.

Competing interests

The author declares no competing financial or non-financial interests.

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